

Lab Report: Calibration and Characterisation of an NTC Thermistor

Course Code & Name: PH 3039 - Data Acquisition Laboratory

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Abstract

An NTC thermistor was calibrated as a temperature sensor, and its response time was measured. The thermistor and a 3.3 k Ω fixed resistor formed a voltage divider read by the 10 bit ADC of an Arduino Uno, and a DS18B20 was the reference, in a water bath heated from 0.3 to 98.5 $^{\circ}\text{C}$ and then cooled. A cubic fit gives $T = 0.00000021266 A^3 - 0.00028857 A^2 + 0.22985 A - 19.007$, with A the ADC count, valid for counts 95 to 851, with an RMSE of 0.226 $^{\circ}\text{C}$ and a 95% prediction half-width of ± 0.44 $^{\circ}\text{C}$ relative to the DS18B20. Heating and cooling differ by at most 0.5 $^{\circ}\text{C}$ on average, so one curve serves both. The two-parameter exponential model gives $\beta = 3992$ K, with an RMSE of 0.43 $^{\circ}\text{C}$.

1. Introduction & Theoretical Background

A sensor is calibrated to ensure proper operation and to find its transfer function, which matters for a thermistor because its output is nonlinear and two parts of one type differ in R_{25} within a tolerance. The resistance of an NTC thermistor follows,

$$R_T = R_{25} \cdot \exp\left(\beta\left(\frac{1}{T} - \frac{1}{298.15}\right)\right)$$

where R_{25} is the nominal resistance at 25 $^{\circ}\text{C}$, T is in kelvin, and β is a material constant. A voltage divider turns the resistance into a voltage that the Arduino reads as an ADC count A from 0 to 1023. The goals were to find $T = f(A)$ and judge the fit; to test the exponential law and find β ; to state the accuracy and precision to measure the response time; and to automate the calibration with a DS18B20.

2. Experimental Setup & Procedure

2.1 Simulations, Apparatus & Materials

Apparatus. An NTC thermistor, a 3.3 k Ω fixed resistor, an Arduino Uno (5 V supply and ADC reference, divider midpoint on pin A2), a DS18B20 at 12 bit resolution on digital pin 2 with a 4.7 k Ω pull up resistor, a PC logging serial output at 9600 baud, a heated water bath, ice, and a room-temperature beaker. Each loop requests a conversion, reads the DS18B20 and the ADC, and logs time, temperature, and ADC count about every 1.5 s. Logging continuously, instead of on demand, means a reading at any time is the nearest row.

Simulation. Both divider arrangements were simulated numerically in Python from 0 to 100 $^{\circ}\text{C}$, for a 3.3 k Ω thermistor and a 3.3 k Ω fixed resistor on 5 V, using the exponential law with $\beta = 3992$ K (found in section 3.2). The two outputs add up to 5 V at every temperature, so their sensitivities are equal in size and opposite in sign (Figure 1), and the question of the most sensitive configuration ends in a tie. For R_{25} of about 9.8 k Ω , the value found for our part, the peak is 49 mV per $^{\circ}\text{C}$ near 43 $^{\circ}\text{C}$ instead of 57 mV per $^{\circ}\text{C}$ near 19 $^{\circ}\text{C}$, which suits a span reaching 98 $^{\circ}\text{C}$ better.

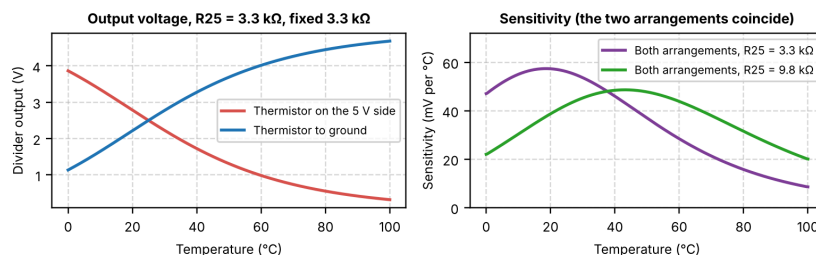


Figure 1. Simulated divider output (left) and its sensitivity (right) from 0 to 100 $^{\circ}\text{C}$.

Configurations. Because the arrangements were tied, both were built, each with its own thermistor, DS18B20, Arduino and PC, and run together in one bath. In configuration 1 (fixed resistor on the 5 V side, thermistor to ground) the first thermistor gave an ADC reading pinned near 1010 to 1022 counts at every temperature, a block of points and not a curve, for which a faulty thermistor or connection is suspected. A replacement on another computer gave a proportional response, but pooling it would mix two devices in one calibration, so this report covers configuration 2 alone.

2.2 Experimental Procedure

Calibration.ipynb. The bath was heated from about 0 to 98.5 $^{\circ}\text{C}$ (log_20261002_141803) and then left to cool (log_20261002_144604). An ice cube added during cooling caused a sudden drop from 84 to 55 $^{\circ}\text{C}$. The 1.5 s interval kept consecutive readings within the 1 $^{\circ}\text{C}$ step limit except in that drop. The DS18B20 was the reference, and its datasheet accuracy was taken as given.

Response.ipynb. The rig was moved by hand between a beaker of room temperature water (about 28 $^{\circ}\text{C}$) and an ice bucket, giving four warming and four cooling steps, of which the three complete cycles in the middle were analysed. We had no second glass for a hot bath, so room temperature water replaced the 80 $^{\circ}\text{C}$ bath, and the logger replaced the voltmeter and phone video.

3. Results & Data Analysis

3.1 Calibration Data Table

The ice drop was cut out by time window, splitting the cooling run into an upper part (85 to 98 °C) and a bottom part (31 to 56 °C), because during a sudden drop the two sensors respond at different speeds and those points show lag, not calibration. Stray points beyond 2.5 standard deviations of a cubic fit were then removed in up to three passes (110 points), leaving 3106.

Data set	N	ADC range	Temperature range (°C)	RMSE (°C)	Largest error (°C)
Heating	1071	95 to 851	0.3 to 98.5	0.345	0.937
Cooling, upper	354	776 to 849	84.9 to 98.4	0.199	0.466
Cooling, bottom	1681	314 to 551	31.5 to 55.7	0.101	0.299
Pooled	3106	95 to 851	0.3 to 98.5	0.226	0.937

Table 1. Cleaned calibration data and the errors of the pooled cubic against the DS18B20, by data set.

3.2 Model Fitting & Parameter Derivation

Polynomial fit. Degrees 1 to 10 were fitted and judged by RMSE, R^2 , adjusted R^2 and residual plots. A straight line leaves a pooled RMSE of about 2 °C and a quadratic barely helps (1.91 °C), but a cubic gives 0.226 °C, whereas degree 10 only reaches 0.172 °C. The remaining scatter is lag and hysteresis, which a higher degree would only fit as noise. Adjusted R^2 equals R^2 to six decimals, so the elbow in RMSE was decided. The pooled cubic is $T = 0.00000021266 A^3 - 0.00028857 A^2 + 0.22985 A - 19.007$, with R^2 of 0.99993.

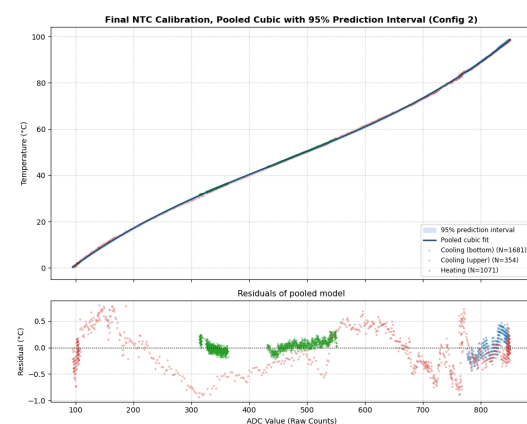


Figure 2. Pooled cubic with its 95% prediction interval (top) and residuals (bottom).

Hysteresis and pooling. Heating and cooling were compared at equal ADC directly from the data, not only from the fitted curves, so that a polynomial could not create or hide a gap. Heating read lower than cooling by 0.17 °C on average over the upper range and 0.49 °C over the bottom range (largest gaps 0.69 and 1.11 °C). This is comparable to the ± 0.5 °C reference accuracy, so the sets were pooled into one cubic with a 95% prediction interval, which gives the range where a single new reading will fall, whereas a confidence interval would shrink to almost nothing with thousands of points. Its half-width is ± 0.44 °C.

Adequacy. The cubic follows the curvature and its residuals stay within about 1 °C (Figure 2): small and smooth for cooling, and a slow wave of 0.5 to 0.9 °C for heating, which we attribute to lag and not to the shape of the sensor, so the fit is adequate.

Calculation of β . With the thermistor on the 5 V side, $A = 1023 \cdot \frac{R_f}{(R_f + R_t)}$, so $r = \frac{R_t}{R_f} = \frac{1023 - A}{A}$, found from the count alone.

The exponential law then gives $\ln(r) = \ln(\frac{R_{25}}{R_f}) + \beta \cdot (\frac{1}{T} - \frac{1}{298.15})$, a straight line in $\frac{1}{T}$ of slope β . Least squares fits give $\beta = 3973$ K (heating), 4026 K (cooling) and 3992 K (pooled), with R^2 of $\ln r$ against $\frac{1}{T}$ of 0.99985, 0.99994 and 0.99976, so β is constant to about 1 per cent (Figure 3). The upper cooling range alone gives 4277 K, but 85 to 98 °C covers too little of $\frac{1}{T}$ to fix β . The intercept gives $R_{25}/R_f = 2.96$, so R_{25} is about 9.8 k Ω for a 3.3 k Ω fixed resistor, probably a 10 k Ω part. In calibration form, $\frac{1}{T} = a \cdot \ln(r) + b$; $a = 0.00025041$ and $b = 0.0030819$ (per K). Its RMSE in temperature is 0.43 °C against 0.23 °C for the cubic, and its residuals form a slow wave of about ± 1 °C (Figure 3, right), so a single β misses some curvature.

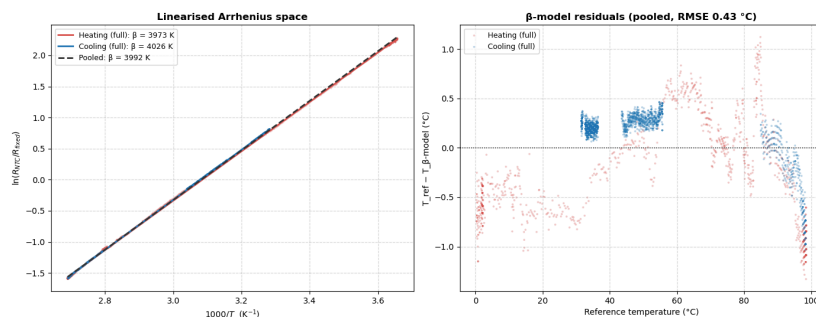


Figure 3. Linearised fit of $\ln r$ against $1/T$ (left) and temperature residuals of the pooled β model (right).

3.3 Transient Response

The ADC was converted to temperature with the pooled cubic. Each warming step was fitted for both sensors separately with a delayed first-order exponential, because the DS18B20 lags too and is reliable only at settled values (Figure 4). The cooling steps are not exponential because the ice contact was uneven, so their 10 to 90 per cent fall time was measured instead, leaving out cycle 2, which had a much sharper dip.

The thermistor is about three times faster than the DS18B20. The ratio of the two time constants (2.7, 3.0 and 3.1) is steadier than either value, because cycle 1 was slow for both sensors, probably from how the rig was plunged. The spread between cycles exceeds the fit standard errors (0.07 to 0.25 s), so it is the honest uncertainty. The DS18B20 rise times match 2.197τ , so one

exponential describes it, while the thermistor rise is 10 to 20 per cent longer. The DS18B20 had not settled when each cooling record ended, so its fall times are lower bounds.

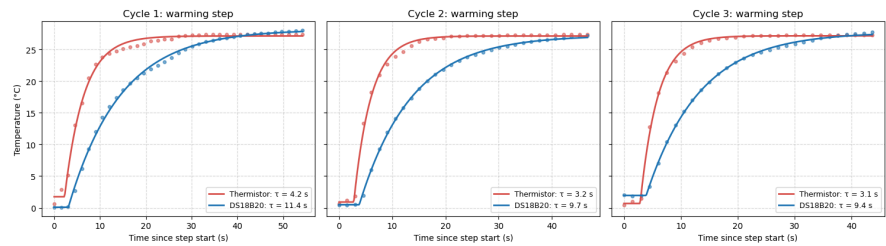


Figure 4. Delayed first order fits to the three warming steps, thermistor (red) and DS18B20 (blue).

Step	Sensor	Time constant τ (s)	10 to 90 percent time (s)
Warming	Thermistor	3.5 ± 0.6	8.8 ± 1.5
Warming	DS18B20	10.2 ± 1.1	22.9 ± 1.8
Cooling	Thermistor	not fitted	18 to 24
Cooling	DS18B20	not fitted	60 to 82 (lower bound)

Table 2. Response times. Warming: mean $\pm$ standard deviation of three cycles. Cooling: range of cycles 1 and 3.

4. Error & Uncertainty Analysis

4.1 Sources of Error

- Measurement precision.** The reference accuracy is a datasheet limit (section 4.2), and one ADC count is 0.10 to 0.20 °C, which sets the resolution.
- Thermal lag and gradient.** The DS18B20 is three times slower than the thermistor, so on a ramp the two sensors disagree (section 3.2). On 5 October, on settled plateaus only, the calibrated thermistor read 1.03 °C below the DS18B20 at room temperature (spread 0.10 °C) but agreed at the ice plateau (difference 0.01 °C, spread 1.0 °C, mostly bath noise). An offset at one end points to a calibration bias, not a shifted rig, and we attribute it to DS18B20 lag during the heating run, which is untested.
- Self-heating.** The divider current dissipates $P = I^2R$ in the thermistor, 1.4 mW at 25 °C and up to 1.9 mW near 50 °C. The temperature rise depends on the dissipation constant, which was not measured.

4.2 Uncertainty Calculations

Standard error of the fit = 0.226 °C for $n = 3106$ points and $p = 4$ parameters. This is the precision, and 1.96 s gives the 95% prediction half-width of ± 0.44 °C. **Error in β .** The datasheet for our part was not available. If it is the common 10 k Ω part with β of 3950 K (an assumption), the pooled 3992 K is 1.1 per cent higher. Hysteresis and ADC quantisation are already inside u_{fit} . The expanded uncertainty U uses a coverage factor of 2 (Table 3).

Condition	u_{fit} (°C)	u_{ref} (°C)	u_{run} (°C)	u_c (°C)	$U, k = 2$ (°C)
Up to 85 °C, same day	0.226	0.289	0	0.367	± 0.73
Up to 85 °C, another day	0.226	0.289	0.635	0.733	± 1.47
Above 85 °C, same day	0.226	1.155	0	1.177	± 2.35
Above 85 °C, another day	0.226	1.155	0.635	1.337	± 2.67

Table 3. Standard and expanded uncertainty of the calibrated temperature against true temperature.

5. Discussion & Conclusion

Sensitivity. In the exponential law, $\alpha = \left(\frac{1}{R}\right) \cdot \left(\frac{dR}{dT}\right)$ equals minus $\frac{\beta}{T^2}$, so it is negative for an NTC, with a size of 5.4 percent per K at 0 °C, 4.5 at 25 °C, 3.8 at 50 °C, 3.1 at 85 °C and 2.9 at 98 °C. In the ADC output, one count is 0.18, 0.12, 0.10, 0.17 and 0.20 °C at those temperatures (from the derivative of the cubic), so the divider is most sensitive near 50 °C, in agreement with the simulation. **Linearity.** T against A is not linear, as a straight line leaves an RMSE of about 2 °C. The slope stays within 25 percent of its minimum (0.10 °C per count) only from about 24 to 67 °C (counts 256 to 649), so the sensor is approximately linear only there. **Practical use.** The cubic needs only a few multiplications on a microcontroller, so the sensor suits monitoring of slow changes.

Conclusion. The final calibration is the pooled cubic of section 3.2, valid for counts 95 to 851 (0.3 to 98.5 °C), with a precision of 0.226 °C that is better than the reference accuracy, because heating and cooling agree to within it and one curve serves both directions. The thermistor follows the exponential law with one constant, $\beta = 3992$ K, but that model has twice the error of the cubic, so curvature matters and the Steinhart-Hart equation is the natural next model. A lag corrected recalibration with checks at the ice and boiling points would test whether lag explains the room plateau offset.